Supplementary Material

Do Domain-Specific Curvature Scales Improve Domain Generalization?
A Controlled Re-evaluation of HyperDG

Anonymous Authors

1 Supplementary Scope and Evidence Status

This supplement separates confirmatory evidence from exploratory or incomplete analyses. This distinction is necessary because the repository contains more artifacts than can support final claims.

Table 1: Evidence inventory at the time of this revision. "Complete" means that all planned configurations reached their configured final step and wrote a checkpoint. "Usable" additionally requires a controlled comparison.

Analysis	Status	Use in paper
Four-condition controlled study	Complete: 256/256	Primary evidence; four seeds and four splits on four datasets.
Component and variance diagnostics	Complete: 36/36	Exploratory seed-0 evidence; not used for population-level claims.
Curvature trajectories	Complete	Mechanistic diagnostic on all four controlled datasets.
Single shared curvature	Partial: 15/16 on four controlled datasets	Seed-0 diagnostic only; TerraIncognita split 1 is excluded.
Poincare projections	Generated	Qualitative post-hoc projections; not evidence of end-to-end hyperbolic training.
Runtime benchmark	Repeated	Reported as an implementation-cost diagnostic, separate from accuracy runs.
MERU comparison	Incomplete	Excluded from accuracy conclusions; four runs lack final checkpoints.
Camelyon17	HyperDG only	Descriptive custom leave-one-hospital-out result; no matched baseline.
λ_{align}/proxy sweep	Confounded	Excluded: available configurations change multiple weights and cover one seed/split.

The primary controlled study is therefore complete. The direct shared-curvature comparison, true SAM comparison, cross-dataset one-factor hyperparameter study, controlled MERU study, and controlled WILDS study are not complete. We do not present these as solved reviewer requests.

2 Expanded Primary Results

2.1 Run inventory

The controlled batches contain 292 completed runs:

- 256 primary runs: four conditions × four datasets × four seeds × four held-out domains;
- 24 seed-0 leave-one-component-out runs: three dropped components × two datasets × four held-out domains;
- 12 seed-0 PACS variance-weight runs: three additional weights × four held-out domains.

Every one of these configurations reaches its expected final step and has a final checkpoint in the local experiment archive. The public git tree contains all 292 configurations and the primary per-run CSV, but not

the raw `experiments/` directory. The manuscript consequently claims release of configurations and aggregate per-run results, not release of every checkpoint.

2.2 All paired effects

Table 2 expands the forest plot in the main paper. Accuracy differences are variant minus ZeroPenalty, paired on seed and held-out domain. Confidence intervals are unadjusted; p-values are shown both raw and after Holm correction over all 12 comparisons.

Table 2: All planned paired comparisons ($n = 16$ each). No comparison remains significant after family-wise error correction.

Variant	Dataset	Δ (pp)	95% CI	Raw p	Holm p
Full	PACS	+0.454	[−1.065, 1.973]	.534	1.000
Full	VLCS	−0.154	[−1.033, 0.724]	.713	1.000
Full	OfficeHome	−1.022	[−1.794, −0.251]	.013	.153
Full	TerraIncognita	−0.733	[−3.921, 2.455]	.631	1.000
Frozen	PACS	+0.626	[−0.620, 1.872]	.301	1.000
Frozen	VLCS	+0.644	[−0.471, 1.759]	.237	1.000
Frozen	OfficeHome	−0.924	[−1.682, −0.167]	.020	.200
Frozen	TerraIncognita	+0.404	[−2.682, 3.491]	.784	1.000
No variance	PACS	+0.529	[−0.837, 1.894]	.422	1.000
No variance	VLCS	−0.586	[−1.876, 0.704]	.348	1.000
No variance	OfficeHome	−1.027	[−1.841, −0.213]	.017	.185
No variance	TerraIncognita	+0.000	[−3.726, 3.727]	1.000	1.000

The TerraIncognita/no-variance result is effectively zero (+0.00036 *pp*, raw $p = .99985$). This value is computed with `scipy.stats.ttest_rel`; the earlier custom incomplete-beta implementation produced an erroneous $p = .0206$ and has been removed.

3 Curvature Dynamics

Each figure compares $\lambda_{\text{var}} = 0$ (left) with the default $\lambda_{\text{var}} = 0.2$ (right). Each panel contains 12 curves: four held-out-domain splits times three source-domain parameters. The curves are positive curvature magnitudes $c_i = \text{softplus}(a_i) + 10^{-4}$, not negative sectional-curvature values. Logged indices refer to source-loader slots; because the held-out domain changes, the same index need not denote the same named domain in every split.

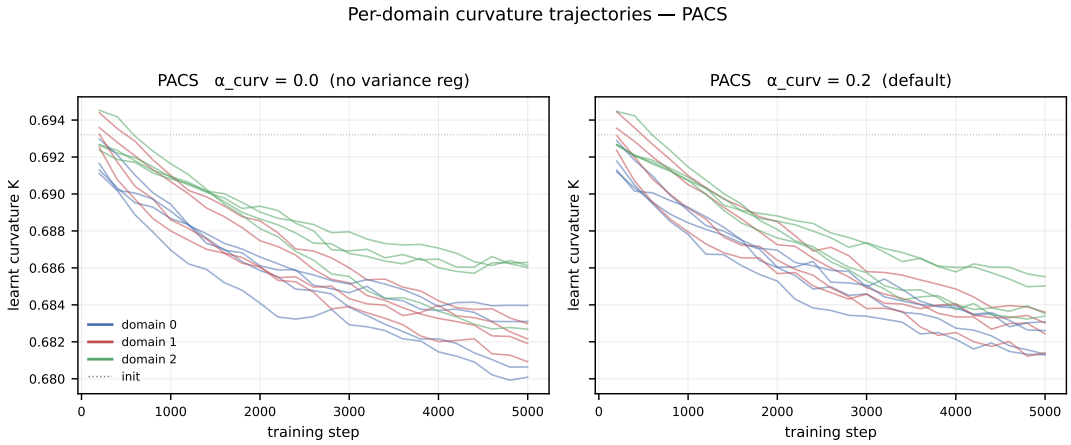


Figure 1: PACS curvature-scale trajectories.

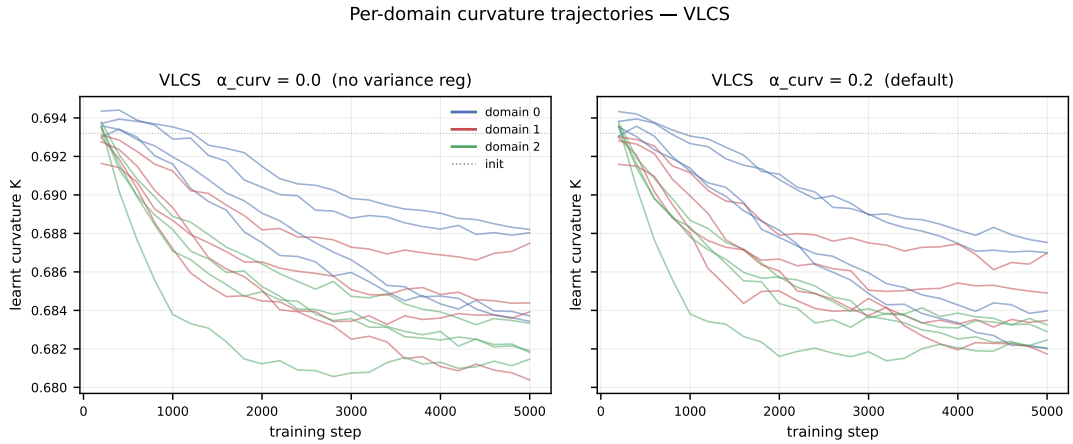


Figure 2: VLCS curvature-scale trajectories.

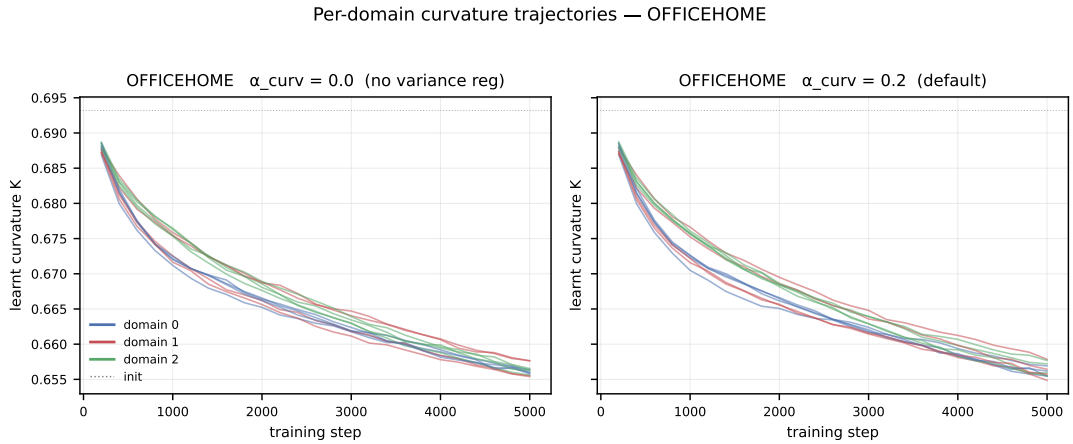


Figure 3: OfficeHome curvature-scale trajectories.

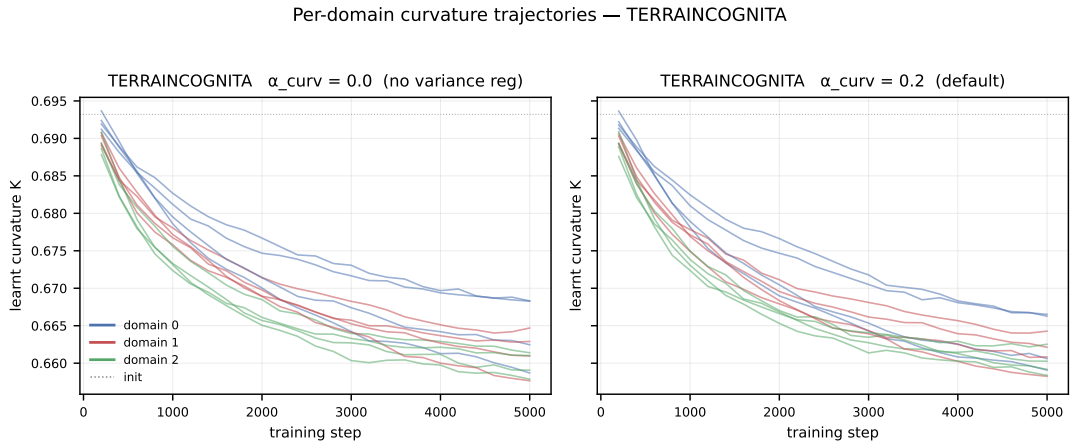


Figure 4: TerraIncognita curvature-scale trajectories.

The visual conclusion is descriptive but consistent across datasets: the parameters follow similar paths with or without the explicit variance penalty. No class-hierarchy metadata are available for these four benchmarks, so we do not claim a correlation between curvature and hierarchy depth.

4 Direct Shared-Curvature Diagnostic

Table 3 compares learned per-domain scales with one learned shared scale. All configurations are seed 0. PACS, VLCS, and OfficeHome have four complete held-out-domain runs. TerraIncognita has three: shared-curvature split 1 stopped at step 2,600 of 5,000 and wrote no final checkpoint, so it is excluded. DomainNet is also excluded because the available per-domain and shared runs use different step budgets and are not a controlled pair.

Table 3: Seed-0 shared-curvature diagnostic. Δ is shared minus per-domain. Standard deviations are across held-out domains, not random seeds.

Dataset	Splits	Per-domain	Shared	Δ (pp)
PACS	4	84.76 ± 7.55	85.29 ± 5.23	$+0.53$
VLCS	4	77.49 ± 15.38	76.21 ± 16.24	-1.28
OfficeHome	4	66.73 ± 10.34	65.50 ± 10.11	-1.23
TerraIncognita	3	44.94 ± 5.53	41.74 ± 1.55	-3.20

These data do not justify equivalence: they use one seed, and TerraIncognita is incomplete. They motivate, but do not replace, a multi-seed shared-curvature condition matched to the primary grid.

5 Component Diagnostics

5.1 Leave-one-component-out analysis

Table 4: Seed-0 leave-one-component-out accuracy (mean $\pm$ sample SD over four held-out domains). Legacy "sharpness" is named norm proxy here because it computes feature-norm variance rather than a SAM perturbation.

Condition	PACS	OfficeHome
Full objective	84.76 ± 7.55	66.73 ± 10.34
Drop domain-mean alignment	84.80 ± 7.91	68.14 ± 9.70
Drop norm proxy	87.06 ± 7.02	66.14 ± 10.90
Drop local/global terms	84.25 ± 8.17	66.70 ± 11.81
Drop curvature variance	83.70 ± 8.22	65.63 ± 10.99
Freeze curvature	85.28 ± 7.48	65.64 ± 11.18
ZeroPenalty	84.09 ± 6.87	67.25 ± 12.02

The apparent $+2.30\,pp$ PACS change after removing the norm proxy does not replicate as a consistent OfficeHome effect and is within the large held-out-domain dispersion. These are exploratory comparisons without multiplicity-adjusted inference.

5.2 Proxy-term decomposition on OfficeHome

The original reviewer request asked for SAM alone versus curvature refinement. The released HyperDG code does not execute SAM. The available 16-run schedule instead toggles the norm-dispersion and local/global proxy terms. We relabel the conditions accordingly in Table 5; calling them SAM would be incorrect.

Table 5: OfficeHome proxy decomposition (seed 0). Each split holds out the corresponding OfficeHome domain.

Condition	Split 0	Split 1	Split 2	Split 3	Mean $\pm$ SD
ZeroPenalty	63.71	54.87	75.20	76.58	67.59 ± 10.26
Proxy terms only	59.79	49.94	75.54	79.10	66.09 ± 13.65
Curvature/alignment only	62.89	50.29	75.87	76.23	66.32 ± 12.36
Full objective	59.79	50.86	74.52	77.73	65.73 ± 12.62

A genuine response to the SAM concern still requires a standard SAM implementation under the same backbone, splits, seeds, and update budget.

5.3 Variance-weight sweep

Table 6: PACS seed-0 sweep of the curvature-variance coefficient. Means and sample SDs are across four held-out domains.

λ_{var}	Accuracy (%)
0.00	83.70 ± 8.22
0.05	83.78 ± 9.10
0.20	84.76 ± 7.55
0.50	85.92 ± 6.09
2.00	86.09 ± 7.63

The table is compatible with a modest effect but is too low-powered to establish stability: it has one seed, and standard deviations across held-out domains exceed differences between means. The existing λ_{align} and legacy feedback/proxy configurations vary several coefficients simultaneously and are therefore not reported as one-factor sensitivity curves.

6 Runtime and Memory

Runtime uses a genuine DomainBed ERM configuration, not the curvature-based `flatland` configuration previously mislabeled as ERM. Each repeat uses 20 warm-up and 200 timed updates on an RTX 6000 Ada, with CUDA synchronization. Method order is rotated across repeats. Peak memory is allocated PyTorch memory, not total device residency.

Table 7: Repeated PACS training-step benchmark (five repeats). Relative cost is computed against ERM within each repeat and then averaged.

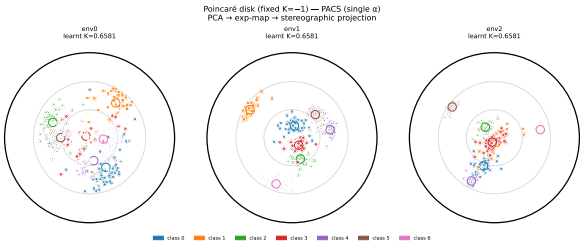
Method	ms/step	Peak allocated MiB	Relative to ERM
ERM	456.16 ± 26.89	8323.9 ± 3.7	1.000 ± 0.000
HyperDG	464.27 ± 8.04	8292.6 ± 12.8	1.020 ± 0.060
SFT	583.67 ± 4.39	3086.1 ± 4.8	1.283 ± 0.068

HyperDG is 8.11 ms/step slower than ERM on the mean, corresponding to a 2.0% average relative overhead with a 6.0% repeat-to-repeat SD. Peak allocated memory is similar for HyperDG and ERM in this benchmark. Thus, the implementation does not impose a large runtime penalty, but the controlled accuracy study also provides no detected gain with which to offset that cost.

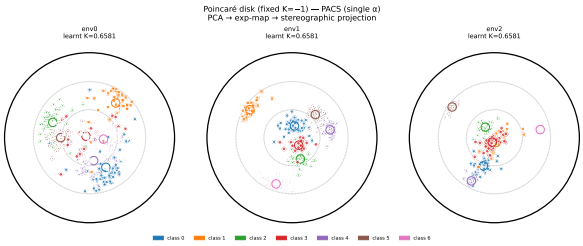
The benchmark measures implementation cost, not end-to-end time including periodic evaluation or checkpoint I/O. Data-loader workers are disabled to reduce host-side variance.

7 Post-hoc Poincaré Projections

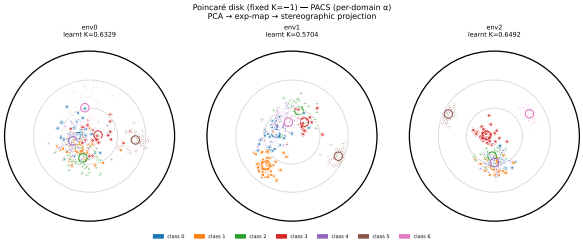
These plots respond to the request for Poincare-disk visualizations, but their interpretation is deliberately limited. For each panel, the plotting script extracts encoder features, divides by the learned scale, fits a separate 2-D PCA, applies robust radial scaling, maps through a fixed-curvature ($K = -1$) exponential map, and stereographically projects to the disk. Because PCA and radial scaling are fitted separately per panel, axes and radial depths are not comparable across panels. The plots show within-panel class organization only; they do not demonstrate learned manifold hierarchy or end-to-end Lorentz training.



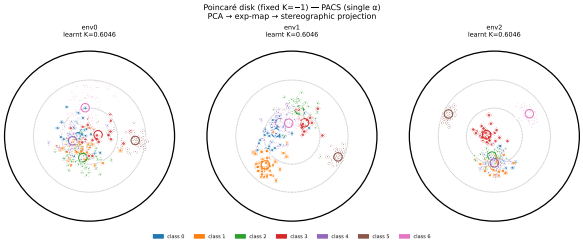
(a) HyperDG, per-domain scale



(b) HyperDG, shared scale

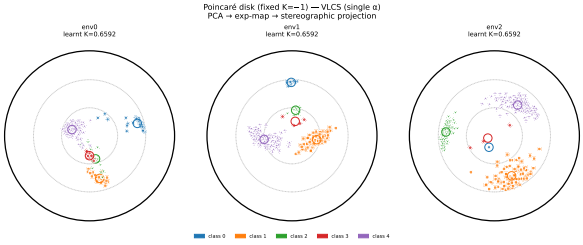


(c) MERU variant, per-domain scale

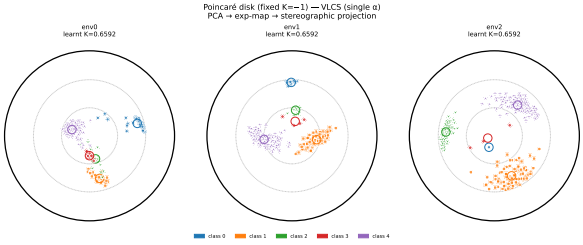


(d) MERU variant, shared scale

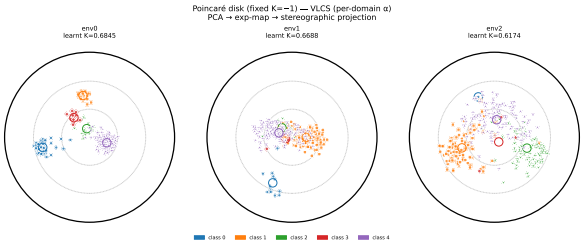
Figure 5: Post-hoc PACS projections (split 0).



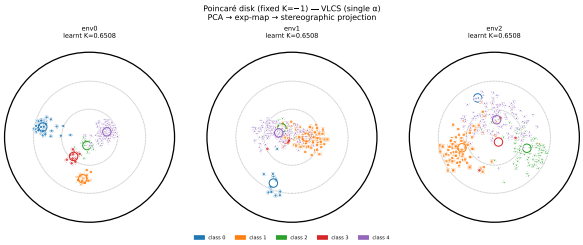
(a) HyperDG, per-domain scale



(b) HyperDG, shared scale

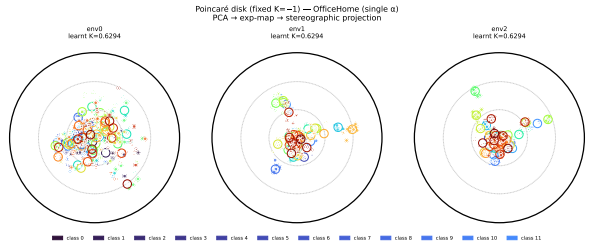


(c) MERU variant, per-domain scale

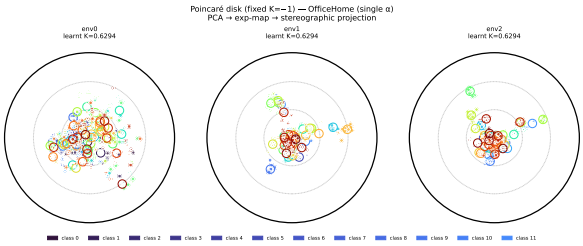


(d) MERU variant, shared scale

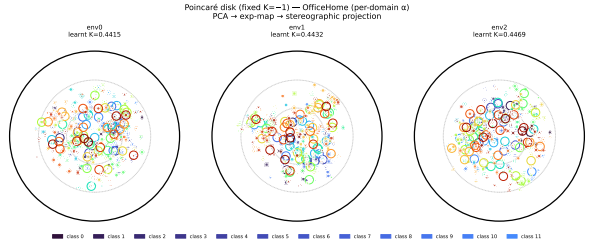
Figure 6: Post-hoc VLCS projections (split 0).



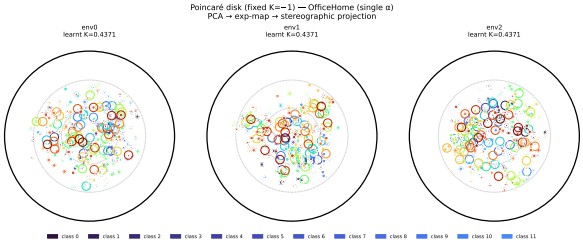
(a) HyperDG, per-domain scale



(b) HyperDG, shared scale

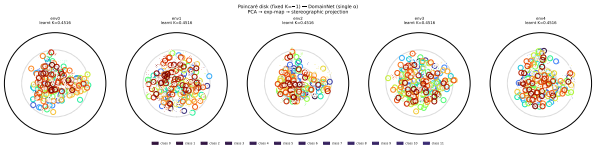


(c) MERU variant, per-domain scale

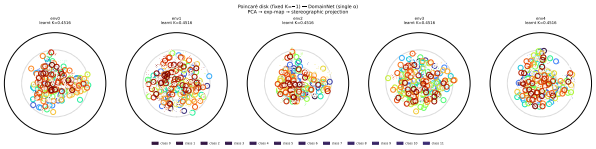


(d) MERU variant, shared scale

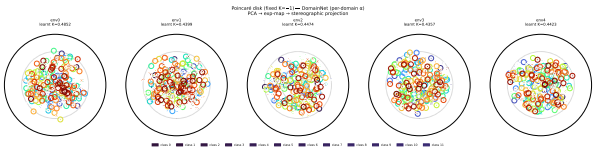
Figure 7: Post-hoc OfficeHome projections (split 0).



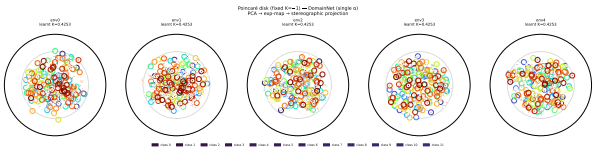
(a) HyperDG, per-domain scale



(b) HyperDG, shared scale

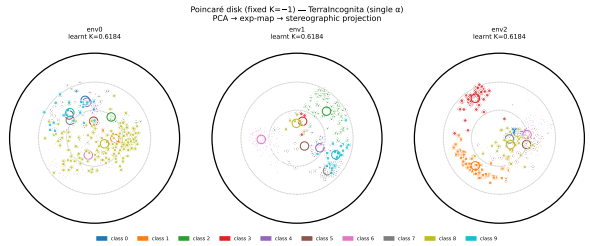


(c) MERU variant, per-domain scale

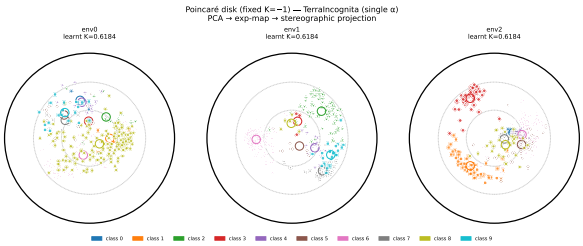


(d) MERU variant, shared scale

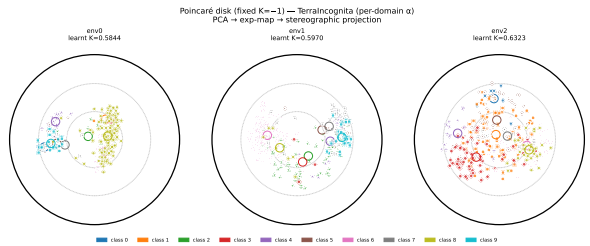
Figure 8: Post-hoc DomainNet projections (split 0). These are visualization artifacts only; the available accuracy runs are not a controlled comparison.



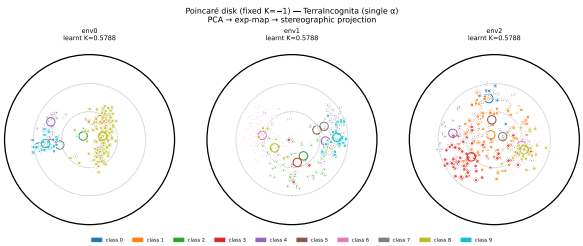
(a) HyperDG, per-domain scale



(b) HyperDG, shared scale



(c) MERU variant, per-domain scale



(d) MERU variant, shared scale

Figure 9: Post-hoc TerraIncognita projections (split 0).

8 Exploratory Benchmarks Not Used for Claims

8.1 MERU status

The repository contains 22 per-domain and 22 shared-scale MERU configurations, all at seed 0. Four did not complete:

- per-domain DomainNet split 4: final recorded step 3,000/4,999;
- shared DomainNet split 3: step 2,000/4,999;
- shared DomainNet split 4: malformed/incomplete final record;
- shared DomainNet split 5: step 4,200/4,999.

Furthermore, the implemented contrastive positive mask selects all same-class, different-index samples after concatenating domains. It does not retain domain IDs and therefore mixes intra-domain and inter-domain positives. The existing runs cannot answer the requested inter-versus-intra contrastive question. The MERU total-loss expression also adds the global consistency term twice; that must be corrected before rerunning. We exclude their accuracy averages rather than presenting an incomplete table.

8.2 Camelyon17 status

Five HyperDG-only runs use a custom leave-one-hospital-out adaptation of Camelyon17. They complete at step 4,999 and yield the descriptive values in Table 8. This is not the official WILDS train/validation/test protocol, has one seed, and lacks a matched ERM or SAM control. It is therefore not evidence that HyperDG improves robustness on WILDS.

Table 8: Exploratory HyperDG-only Camelyon17 leave-one-hospital-out accuracy. The final row is sample SD over hospitals.

Held-out hospital	Accuracy (%)
0	97.03
1	93.34
2	95.67
3	93.43
4	63.86
Mean	88.67
SD	13.95

9 Reproduction Commands

The principal generated artifacts are reproduced with:

```
PYTHONPATH=third_party/domainbed conda run -n sftdg \
  python scripts/aggregate_batch4.py
PYTHONPATH=third_party/domainbed conda run -n sftdg \
  python scripts/plot_forest_erm_vs_hyperdg.py
PYTHONPATH=third_party/domainbed conda run -n sftdg \
  python scripts/plot_curvature_trajectories.py
PYTHONPATH=third_party/domainbed conda run -n sftdg \
  python scripts/aggregate_secondary.py
```

The main aggregation writes raw per-run accuracy and effect summaries to `outputs/batch4_per_run.csv` and `outputs/batch4_summary.csv`. Secondary tables are generated as separate CSV files under `outputs/`.